

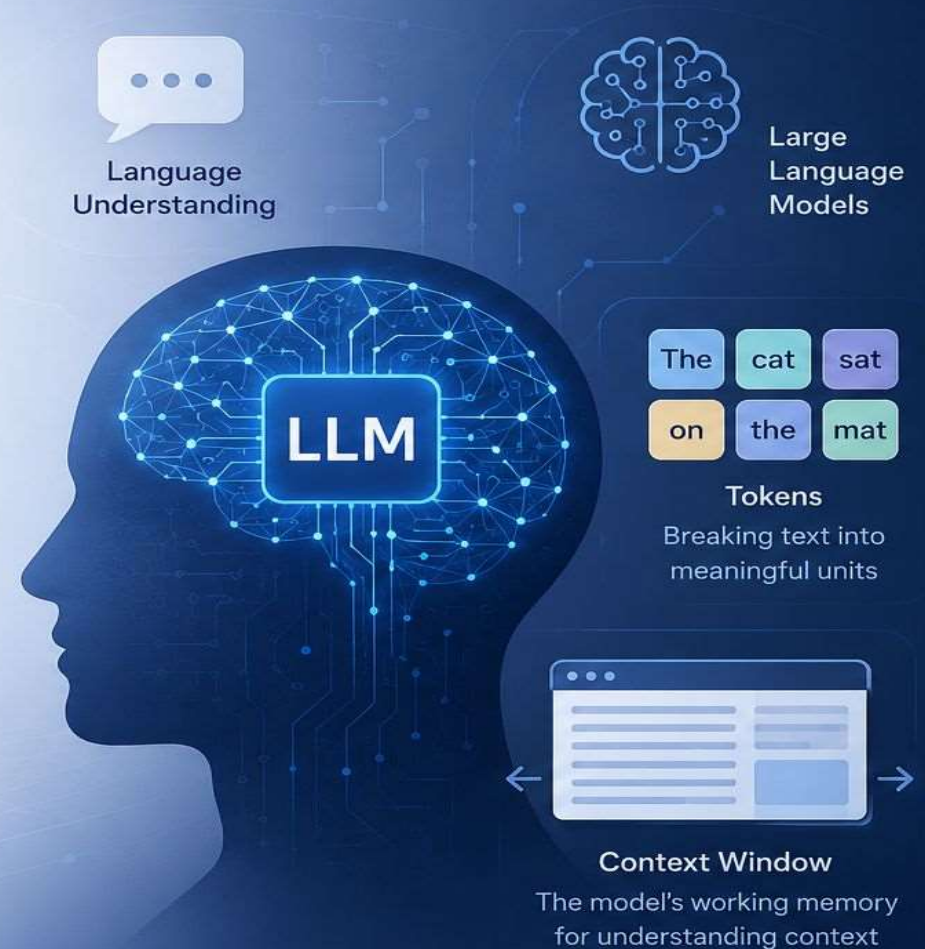
LLM Fundamentals

The Building Blocks of
Intelligent Language Systems



Introduction to LLMs, Tokens & Context Window

Understanding the building blocks of Large Language Models



What is an LLM?



LLM (Large Language Model) is an AI model trained on massive amounts of text data to understand and generate human-like language.

Examples:



ChatGPT
(OpenAI)



Claude
(Anthropic)



Gemini
(Google)

Real-Life Example

Input:



Explain Python loops



Output:

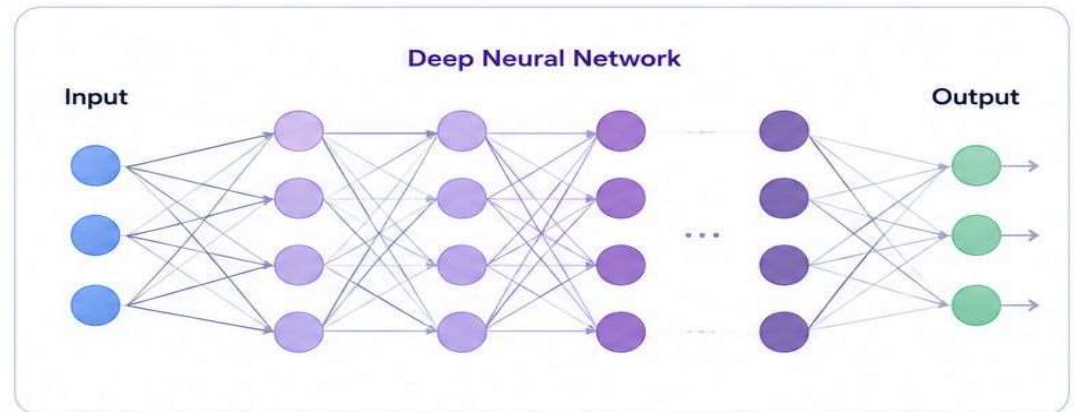


A loop is used to execute a block of code repeatedly.



The LLM predicts the next most likely word based on previous words.

How LLMs Work



Example



Input:

The capital of India is



LLM predicts:

New Delhi



because it has learned patterns from training data.

What are Tokens?



LLMs do not understand words directly.



They break text into **tokens**.



Text Input



LLM



101

2056

309

5021

...

Tokens



Tokens can be words, parts of words, or even punctuation.

Example 1

Text:

Hello World



Tokens:

Hello

World



2 Tokens

Example 2

Text:

Artificial Intelligence



Tokens:

Artificial

Intelligence



2 Tokens

Example 3

Text:

I love Mysore.



Tokens:

I

love

Mysore

.



4 Tokens



Note: Different LLMs may tokenize the same text differently depending on their tokenizer and vocabulary.

Why Tokens Matter



AI cost is calculated based on **tokens**.

Example:

 Input Tokens

+

 Output Tokens

=

 Total Cost

If:



Input

=

500 tokens



Output

=

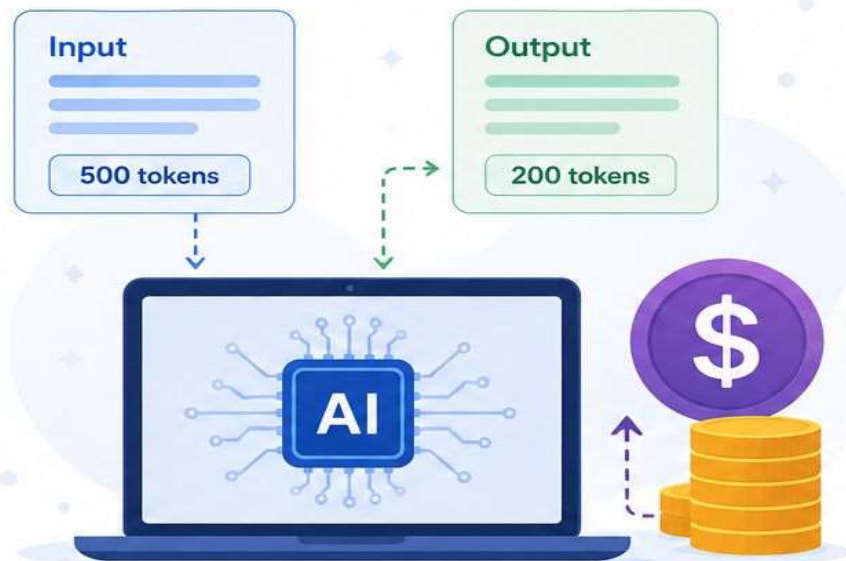
200 tokens



Total

=

700 tokens



Total Cost
(Based on Tokens)

=

700
tokens



More tokens used = Higher cost. Optimize your prompts and responses to manage token usage efficiently.

Context Window



Definition

The context window is the amount of information an LLM can remember during a conversation.

Think of it as the AI's working memory.



Human Analogy

Teacher asks:

"What did I explain 5 minutes ago?"

If student remembers:



Good Context

If student forgets:



Context Lost



LLMs behave similarly.



Example

Prompt:

My name is Naveen.
I am a Technical Trainer.



Later:

What is my profession?



Answer:

Technical Trainer

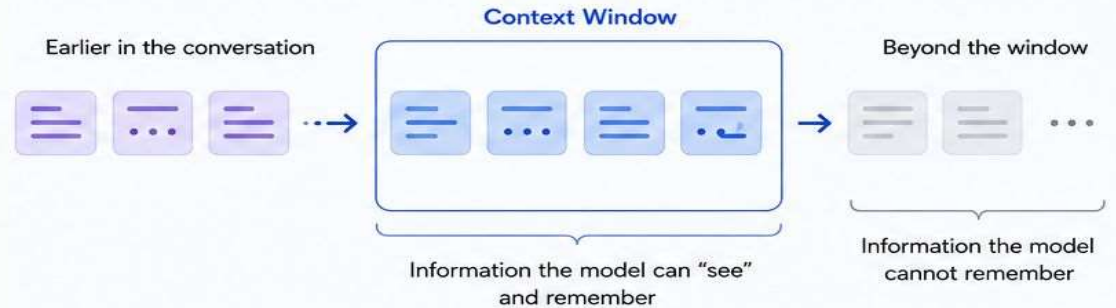


The model uses information from the context window.



Key Takeaway: A larger context window helps the LLM remember more, understand better, and provide more accurate responses.

How Context Window Works



Context Window Sizes

	Model	Approx Context
	GPT-3.5	16K Tokens
	GPT-4	128K Tokens
	Claude	200K+ Tokens
	Gemini	1M+ Tokens

Larger context allows:



Bigger documents



Long conversations



Better memory

Interview Questions



Q1. What is an LLM?

A Large Language Model trained on huge text datasets to understand and generate human language.



Q2. What is a Token?

The smallest unit of text processed by an LLM.



Q3. Why are Tokens Important?

- ✓ Cost calculation
- ✓ Context management
- ✓ Performance measurement



Q4. What is Context Window?

The maximum amount of text an LLM can process and remember at one time.



Key Takeaway: Understanding these basics helps you work effectively with LLMs and build powerful AI applications.



Hands-On Activity



Open ChatGPT and try:

“ My name is Naveen.
I work as a Technical Trainer in Infosys Mysore.
I am learning LangGraph.



Then ask:

“ Where do I work?



Observe how the model uses context.



The model looks at the earlier information within the context window and answers correctly: **“You work in Infosys Mysore.”**